

Practice for Quiz2: MA 5104, Department of Mathematics, IIT Madras

Course name : <i>Statistical Inference and Learning</i>

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1. A bank's credit-scoring system approves a loan application. Approval means the money is disbursed; rejection means the applicant is turned away.
 - (a) What hypotheses are being tested if approving the loan is a type I error?
 - (b) What hypotheses are being tested if approving the loan is a type II error?
 2. A supplier claims that 40% of its orders arrive on time. In a random sample of 18 orders, the hypothesis $p = 0.4$ is rejected in favour of $p < 0.4$ if 3 or fewer arrive on time. Use the binomial distribution.
 - (a) Find the probability of committing a type I error.
 - (b) Find the probability of committing a type II error for the alternatives $p = 0.2$ and $p = 0.1$.
 3. A process has known standard deviation $\sigma = 6$. A sample of $n = 36$ is used to test $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$, and H_0 is rejected if $|\bar{X} - 50| > 2.5$.
 - (a) Find the probability of committing a type I error.
 - (b) Find the probability of committing a type II error when $\mu = 52$.
 - (c) With $\alpha = 0.05$ instead, what sample size is needed for the power at $\mu = 52$ to be at least 0.90?
 4. A random sample of 20 items from a normal population gives $\bar{x} = 8.6$ and $s = 1.2$. Test $H_0 : \mu = 8$ against $H_1 : \mu > 8$ at the 5% level. Note $t_{0.05, 19} = 1.729$. State the hypotheses, the test statistic and its null distribution with degrees of freedom, the rejection region, your decision, and your conclusion in context.
 5. Two independent random samples are drawn from normal populations with equal variances:

Sample 1	$n_1 = 10$	$\bar{x}_1 = 42.5$	$s_1^2 = 6.2$
Sample 2	$n_2 = 12$	$\bar{x}_2 = 39.8$	$s_2^2 = 5.4$

Test $H_0 : \mu_1 = \mu_2$ against a two-sided alternative at the 5% level. Note $t_{0.025, 20} = 2.086$. Show the pooled variance explicitly.

6. Ten subjects are measured before and after a training programme. The differences (after – before) have mean $\bar{d} = 2.3$ and standard deviation $s_d = 3.1$. Test $H_0 : \mu_d = 0$ against $H_1 : \mu_d > 0$ at the 5% level, with $t_{0.05, 9} = 1.833$. State also, in one sentence, why a two-sample test would be inappropriate for these data.

7. A process is regarded as acceptable if the variance of its output is at most 8. A random sample of $n = 25$ items gives $s^2 = 12.4$. Test $H_0 : \sigma^2 = 8$ against $H_1 : \sigma^2 > 8$ at the 5% level, with $\chi_{0.05, 24}^2 = 36.415$. Comment on how close the decision is.

8. A random sample of $n = 15$ from a normal population gives $s^2 = 0.36$.

(a) Test $H_0 : \sigma^2 = 0.25$ against $H_1 : \sigma^2 \neq 0.25$ at the 5% level, using $\chi_{0.025, 14}^2 = 26.119$ and $\chi_{0.975, 14}^2 = 5.629$.

(b) Construct a 95% confidence interval for σ^2 .

(c) Verify that your answers to (a) and (b) are consistent, and explain in one sentence why they must be.

9. Using the data of Question 5, test at the 5% level whether the assumption of equal variances made there is justified. Use $F_{0.025, 9, 11} = 3.588$ and $F_{0.975, 9, 11} = 0.256$. Comment on how much reassurance this test provides.

10. A die is rolled 120 times and the six faces occur 15, 23, 18, 25, 21, 18 times respectively. Test at the 5% level whether the die is fair, using $\chi_{0.05, 5}^2 = 11.070$. State the hypotheses, the expected counts, the statistic with its degrees of freedom, and your conclusion.

11. A genetic model predicts that four phenotypes occur in the ratio 9 : 3 : 3 : 1. Among 160 offspring the observed counts are 95, 28, 24, 13. Test the model at the 5% level, with

$\chi^2_{0.05,3} = 7.815$. Explain why the degrees of freedom are not reduced by an estimated parameter here.

12. Items produced on three shifts are classified as defective or acceptable:

	Defective	Acceptable
Morning	20	180
Afternoon	30	170
Night	50	150

Test at the 5% level whether shift and quality are independent, using $\chi^2_{0.05,2} = 5.991$. State the expected counts, the statistic and its degrees of freedom, your decision, and identify which shift contributes most to the statistic.

13. A sample of $n = 15$ repair costs has sample median 4,200. From $B = 2000$ bootstrap resamples, the ordered bootstrap medians have 50th value 3,150 and 1950th value 5,600.

- (a) Give the 95% bootstrap percentile interval for the population median.
- (b) State in one sentence what the bootstrap distribution is approximating, and in one sentence what it is not.
- (c) Explain why a formula-based interval was not used, and what would have been required to construct one.
- (d) The analyst repeats the procedure with $B = 50,000$ and obtains (3,190, 5,570). Is this interval more accurate than the first? Distinguish carefully between the two sources of error involved.

14. In 20 independent trials of a process, 7 are successes. Let p denote the success probability, with prior $p \sim \text{Beta}(2, 2)$.

- (a) Derive the posterior distribution of p .

- (b) Give the posterior mean and the posterior mode, and compare both with the maximum likelihood estimate.
 - (c) The 95% equal-tailed credible interval is (0.197, 0.573). State precisely what this asserts, and then state what a 95% confidence interval would assert. The two statements must not be interchangeable.
 - (d) State one situation in which the choice of prior would visibly change the interval.
15. In a survey of 300 M.Tech. students, 120 reported using a particular software package; among 280 M.Sc. students, 90 reported the same. Test at the 5% level whether the proportion of users differs between the two programmes.
- (a) State the hypotheses, the test statistic and its null distribution, and the rejection region.
 - (b) Compute the pooled proportion and the observed test statistic, and state your decision.
 - (c) Explain in one sentence why the standard error is computed using a *pooled* estimate of the common proportion rather than the two separate sample proportions.
16. Before a process change, 45 of 500 items were defective; after the change, 28 of 450 were defective. Test at the 5% level whether the change has reduced the defect rate. State the hypotheses, the statistic, the rejection region, your decision, and your conclusion in context.
17. Using the data of Question 15, construct a 95% confidence interval for $p_1 - p_2$.
- (a) Compute the interval, using the *unpooled* standard error.
 - (b) State whether the interval is consistent with your decision in Question 13, and comment on how close the case is.
 - (c) Explain why the pooled standard error is used for the test but the unpooled one for the interval.
18. Two independent random samples from normal populations give

$$\text{Sample 1 } n_1 = 12 \quad \bar{x}_1 = 85.4 \quad s_1^2 = 24.5$$

$$\text{Sample 2 } n_2 = 15 \quad \bar{x}_2 = 79.1 \quad s_2^2 = 61.3$$

- (a) Test the equality of the two variances at the 5% level, using $F_{0.025, 14, 11} = 3.359$.
- (b) In the light of (a), carry out an appropriate test of $H_0 : \mu_1 = \mu_2$ against a two-sided alternative at the 5% level. If you use the Welch procedure, compute the approximate degrees of freedom; take $t_{0.025, 24} = 2.064$.
- (c) Repeat the test assuming equal variances, and comment on whether the two procedures lead to the same conclusion here.

19. Two large independent samples of service times give

$$\text{Branch A } n_1 = 150 \quad \bar{x}_1 = 3.42 \quad s_1 = 0.85$$

$$\text{Branch B } n_2 = 180 \quad \bar{x}_2 = 3.19 \quad s_2 = 0.92$$

- (a) Test $H_0 : \mu_1 = \mu_2$ against a two-sided alternative at the 5% level.
- (b) Give a 95% confidence interval for $\mu_1 - \mu_2$.
- (c) The service-time distributions are strongly right-skewed. Explain in one sentence why this does not invalidate the procedure here, and state what would have been the concern had n_1 and n_2 been 10 and 12.

20. Two hundred customers are classified by whether they were offered a discount and whether they completed a purchase:

	Purchased	Did not purchase
Discount	38	62
No discount	57	43

Test at the 5% level whether offering a discount and completing a purchase are independent, using $\chi_{0.05, 1}^2 = 3.841$. State the expected counts, the statistic and its degrees of freedom, and

your decision. Comment on the direction of the association, and on whether the test alone licenses a causal claim.

21. Three hundred respondents are cross-classified by age group and preferred mode of commute:

	Two-wheeler	Public transport	Car
Under 25	52	38	30
25–40	28	42	35
Over 40	20	30	25

- (a) State the hypotheses and compute the table of expected counts.
- (b) Compute the test statistic, state the degrees of freedom, and test at the 5% level using $\chi^2_{0.05, 4} = 9.488$.
- (c) The decision is a marginal one. State the smallest significance level at which you would reject, to the nearest percent, and comment on how you would report this result.
- (d) Verify that the χ^2 approximation is adequate.